

June 22, 2026

Dr. Joseph Lifton
Lecturer in Mechatronics
School of Engineering
University of Southampton

Dear Dr. Joseph Lifton,

I am writing to express my strong interest in the PhD position focusing on AI-enhanced X-ray CT for defect detection in advanced manufacturing. As an AI & Data Science Engineer with a specialized focus on applying deep learning to complex physical systems, the challenge of automating 3D defect characterization aligns perfectly with my research trajectory.

During my State Engineer thesis, "Gas Turbine Digital Twin," I engineered predictive maintenance architectures to forecast thermodynamic degradation and sensor drift in advanced industrial machinery. By utilizing Spatial-Temporal Transformers (ST-KDFormer) and Temporal Convolutional Networks, I successfully built systems that identify anomalous sub-surface mechanical behaviors from high-dimensional SCADA telemetry without relying on brittle, rule-based heuristics.

Furthermore, during my Master's thesis on the "Cardiovascular Medical Digital Twin," I worked extensively with complex medical imaging and volumetric representations. I developed PI-DeepONet, a Physics-Informed Operator Network that inherently understands the spatial and temporal continuity of complex systems via PyTorch Autograd. I believe these experiences—transitioning from rule-based engineering models to robust, high-dimensional neural networks—perfectly equip me to transition X-ray CT defect analysis from manual, parameter-heavy workflows to fully automated, 3D AI-driven pipelines.

I possess strong programming skills in Python (PyTorch, TensorFlow) and am highly experienced in bridging the gap between computer vision and advanced manufacturing requirements.

Please find attached my CV, references, and a detailed summary of my engineering theses. I would welcome the opportunity to discuss how my background could contribute to your research team.

Sincerely,
Ismail Aissaoui
<https://ismail-aissaoui.vercel.app>